

ISLA MAGUEYES, Puerto Rico

WMO: 997350

Lat: 17.9700N

Lon: 67.0464W

Elev: 5

StdP: 101.26

Time Zone: -4.00 (W04)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	20.5	21.1	N/A	N/A	N/A	N/A	N/A	N/A	8.0	27.2	6.6	27.0	1.5	50	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.3	31.4	N/A	31.1	N/A	30.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	2.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.4	4.9	19.2	32.4	0.6	0.6	18.8	32.9	18.4	33.2	18.1	33.5	17.6	33.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.7	25.2	25.0	25.0	26.0	27.0	28.0	28.0	28.1	28.0	27.7	26.8	25.8		
	DBStd	1.48	0.77	0.86	1.12	0.97	0.90	0.63	0.70	0.74	0.77	0.83	0.88	0.98		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	6105	470	421	466	479	527	539	559	561	540	550	503	490		
	CDD18.3	3063	212	188	207	229	268	289	301	303	290	292	253	232		
	CDH23.3	30601	1539	1389	1558	2043	2774	3392	3522	3574	3326	3211	2384	1888		
	CDH26.7	8759	242	193	207	408	739	1144	1246	1287	1154	1058	659	423		
(14) Wind	WSAvg	2.0	1.8	2.0	2.2	2.3	2.1	2.1	2.0	2.0	1.9	1.9	1.7	1.8		
(15) Precipitation	PrecAvg	1537	85	62	71	123	155	113	132	167	189	176	164	101		
	PrecMax	2298	199	136	149	310	301	238	270	310	450	432	441	227		
	PrecMin	836	29	30	7	24	63	38	61	76	59	75	31	30		
	PrecStd	379	43	26	37	72	73	51	48	45	92	84	80	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.0	29.5	30.0	31.0	31.4	31.7	31.7	31.7	31.7	30.8	30.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	28.7	28.5	28.8	29.4	30.2	30.9	31.2	31.2	31.2	31.1	30.4	29.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	28.4	28.2	28.3	29.0	29.7	30.6	30.8	30.9	30.8	30.7	30.0	29.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	28.0	27.9	27.8	28.6	29.4	30.2	30.5	30.6	30.5	30.3	29.6	28.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	5.9	5.5	5.2	4.8	4.7	5.2	5.3	5.3	5.3	5.8	6.2		
		MCDBR	6.3	6.0	5.5	5.3	4.9	5.0	5.4	5.4	5.4	5.6	6.0	6.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.372	0.392	0.389	0.417	0.457	0.530	0.524	0.488	0.436	0.391	0.386	0.382		
	taud		2.551	2.480	2.493	2.435	2.337	2.092	2.109	2.241	2.421	2.558	2.572	2.554		
	Ebn at Noon		914	911	922	891	845	777	784	819	865	900	893	892		
	Edh at Noon		97	109	111	118	129	163	161	142	118	100	95	95		
(44) All-Sky Solar Radiation	RadAvg		4.79	5.56	6.19	6.48	6.25	6.28	6.36	6.24	5.86	5.25	4.63	4.50		
	RadStd		0.15	0.12	0.23	0.31	0.40	0.34	0.29	0.23	0.41	0.38	0.35	0.19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+0.33	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page